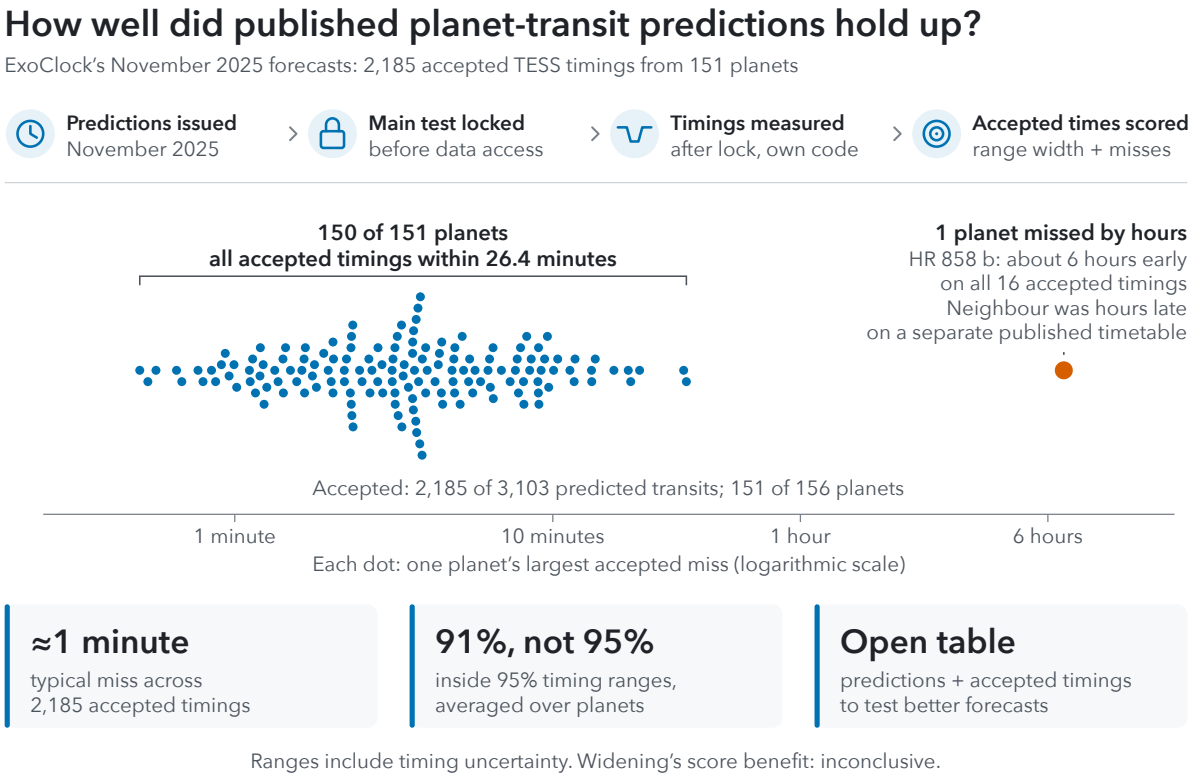
Predicted planet transits arrived within minutes, with one six-hour miss

Research brief · 12 November 2026 · aligned with the reconstructed article (18_final_draft.md). Codes in brackets are claim IDs in the project's claims ledger.



In one sentence: Published predictions of when planets pass in front of their stars agreed with later measurements to minutes; one of 151 tested planets missed by hours.

Why it matters

During a transit, a planet crosses its star. Starlight passing through the planet's atmosphere can reveal its composition. JWST and the coming Ariel mission schedule observations from predicted times and uncertainties: too small, and the transit can be missed; too large, and telescope time is wasted.

ExoClock publishes dated predictions for Ariel. Its own check of the 2025 edition was partly optimistic: some comparison timings had helped build it. Among the studies we checked, none fixed an edition in advance, measured later transits independently and scored its uncertainties this way.

What we did

We measured mid-times, the centre of each crossing, from later NASA TESS data, locking the main test before access (C042, C045). Accepted artificial transits fell within our ranges at about the stated rate, on the samples used to tune the method (C038). We tested a correction learned from the previous edition's errors. The table and scoring package let anyone test new forecasts against these accepted timings (C053).

What we found

Accurate to minutes. We accepted 2,185 timings of 151 of the 156 planned planets. The typical miss was about a minute, and for 150 planets every accepted timing came within 26.4 minutes of its prediction (C043, C063). The 2022 edition still held within 37 minutes nearly four years on, for every planet we could check (C062).

Ranges a little too narrow. Ranges labelled "95%", combined with our measurement uncertainty, contained about 91% of accepted timings, averaging equally across planets (C063). Misses were more often early than late, for reasons we have not identified (C065).

A correction did not clearly help. Widening the ranges 1.7-fold raised coverage above 95%, but the overall score could have improved or worsened substantially (C043). Of 151 planets, 111 scored worse; most of the average gain came from one planet, whose transits even the widened ranges never contained (C046, C059).

One six-hour miss. HR 858 b arrived about six hours early on all 16 accepted timings, as two independently written programs confirm. Against separate published timetables, its neighbour c ran nine to ten hours late while d was on time: the pattern expected when planets tug on each other, and evidence against a shared clock shift (C048, C050, C051).

Warning signs, unproven. HR 858 b was new to the edition, its prediction rested on poorly fitting observations that ended in 2020, and its timing variations had been published a year earlier (C048, C054). But one case cannot validate a warning rule, and on the 2022 edition no tested scheme for giving selected planets wider windows captured more measured midpoints than both ways of spreading the same extra time evenly (C061, C064). This test, the four-year comparison and the early/late check were added after the data had been inspected.

What it means

For nearly every planet tested, an ExoClock prediction agreed with the measured transit to minutes, but the rare exception can be large. Warning information such as old inputs, a poor fit or published timing variations could be printed beside each prediction and tested in future editions. Publishing how the uncertainties of a reference transit time and the interval between transits are linked would make such tests more exact; we had to estimate that link (C054, C063).

What we can and cannot say

- **Only accepted timings count:** 70% of those predicted, and 92% of those TESS was watching at the predicted time (C046). A transit far off schedule could be lost or assigned to the wrong orbit, so no catalogue-wide failure rate follows, and a mid-time does not guarantee capturing a whole transit.
- **One test:** one edition over its first year, on stars with one planet listed in that edition. The wider sample yielded timings for only five planets, and identification on multi-planet stars remains unvalidated (C044). Our cross-checks re-used the same TESS data, and one measurement run was repeated after a file-path fault, preserving every earlier result (C045).

The record

Code, results, claims ledger, reviews and author correspondence are in the public repository; the table and scoring package will get a DOI on release.

Sources: ExoClock catalogue editions on OSF; NASA TESS data via MAST; HR 858 timing variations from Bonfanti et al. 2025, doi:10.1051/0004-6361/202451300.

Barry W. Brook set the question, authorised the opening of the protected data and reviews release versions. Claude Opus 5.5 (Anthropic) and Codex GPT-6.1-sol (OpenAI) designed and ran the analyses, each built an independent timing check, and reviewed each other's work. Claude drafted this brief and its graphical abstract; Codex cross-checked both. Code: MIT. Text: CC BY-SA 4.0.